

Areas under multiple curves



1 For each of the following pairs of functions:

- i Graph the functions on the same set of axes.
- ii Hence, find the area of the region bounded by the functions and the x -axis.

a $x + y = 2$ and $y = x$.

b $x + y = 6$ and $y = 2x$.

2 For each of the following pairs of functions:

- i Graph the functions on the same set of axes.
- ii State the value(s) of x at which the two functions intersect.
- iii Hence, find the area of the region bounded by the functions and the x -axis.

a $y = 2x$ and $y = (x - 4)^2$.

b $y = x^2$ and $y = (x - 2)^2$

3 Find the area of the region bounded by the following pairs of functions and the x -axis:

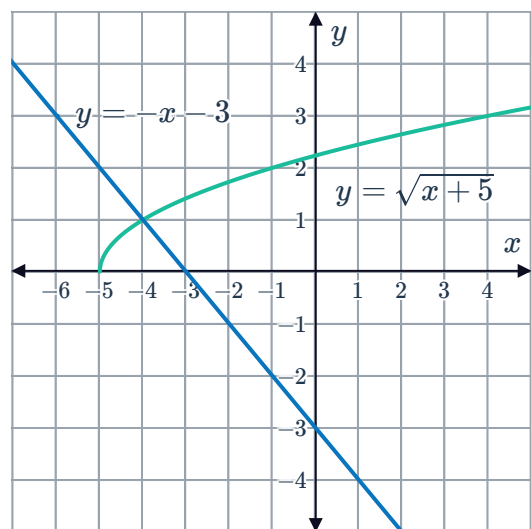
a $y = (x - 1)^2$ and $y = (x + 3)^2$

b $y = x^2 - 4$ and $y = -2x + 11$.

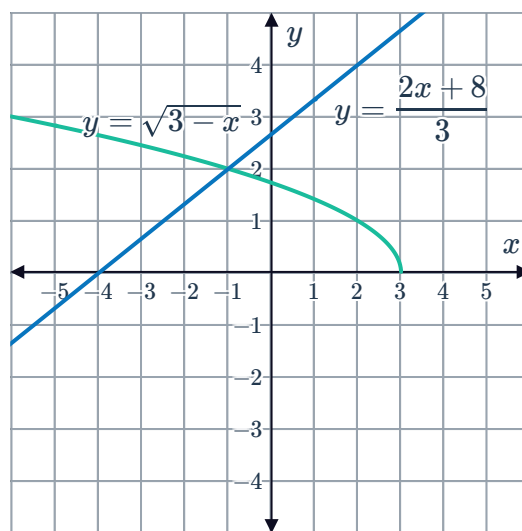
c $y = x^3$ and $y = -3x + 14$.

d $y = (x + 2)^2$ and $y = -4x - 3$.

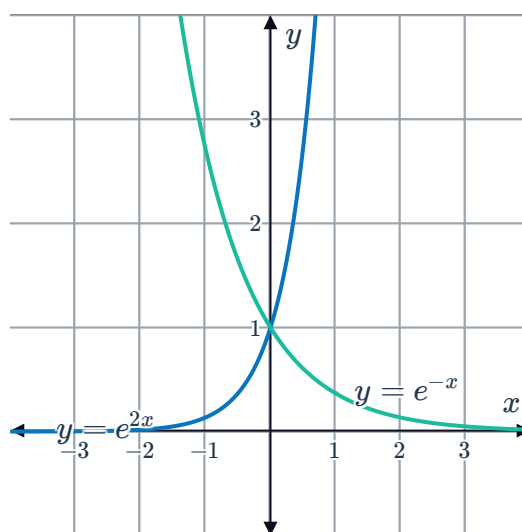
4 Find the area of the region bounded between the curve $y = \sqrt{x + 5}$, the line $y = -x - 3$ and the x -axis:



- 5 Find the area of the region bounded between the curve $y = \sqrt{3-x}$, the line $y = \frac{2x+8}{3}$ and the x -axis:

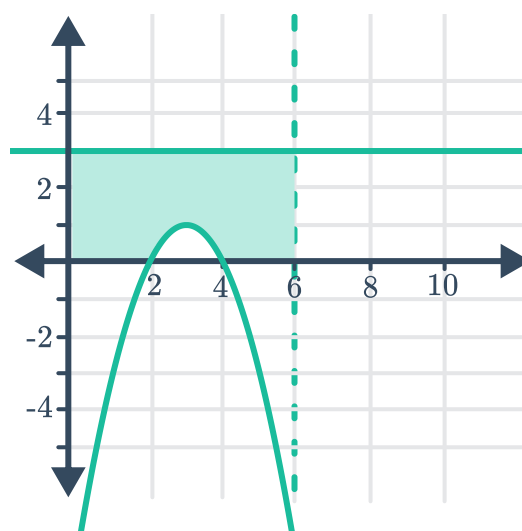


- 6 Find the area of the region bounded by the curves $y = e^{2x}$ and $y = e^{-x}$, the x -axis, and the lines $x = -2$ and $x = 2$:

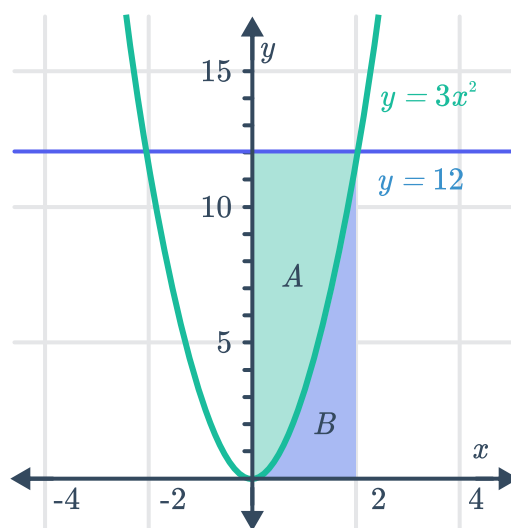


- 7 Find the area of the region bounded by $y = x^2 + 1$, $y = -\frac{x}{2} + 6$, the x -axis, and the y -axis.

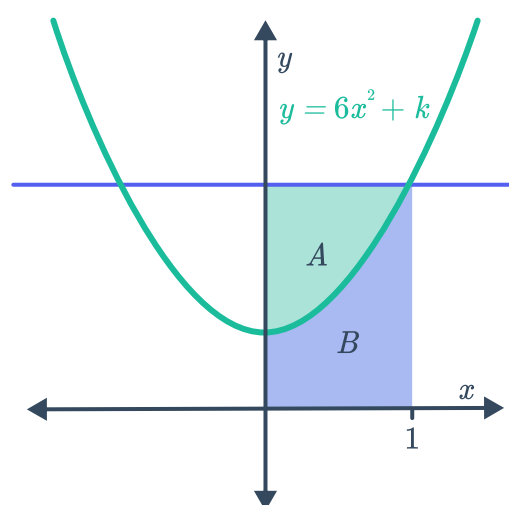
- 8 The diagram shows the shaded region bounded by $y = 3$, $y = 0$, $y = 6x - x^2 - 8$, $x = 0$ and $x = 6$:
Find the area of the shaded region.



- a Find the area of the shaded region B .
- b Find the area of the shaded region A .
- c What is the ratio of the areas $A : B$?



- 10 Consider the following graph:
Find k such that the two shaded regions have equal area.



- 11 Find the exact area between the graph of $y = \sqrt{4 - x^2}$ and the lines $x = 2$ and $y = 2$.

- 12 Consider the functions $y = e^{\frac{x}{6}}$ and $y = e^{\frac{3}{2}}$.

- a Find the x -coordinate of the point of intersection.
- b Calculate the exact area bounded by $y = e^{\frac{x}{6}}$, the y -axis and the line $y = e^{\frac{3}{2}}$.

- 13 Consider the curve $y = e^x$.

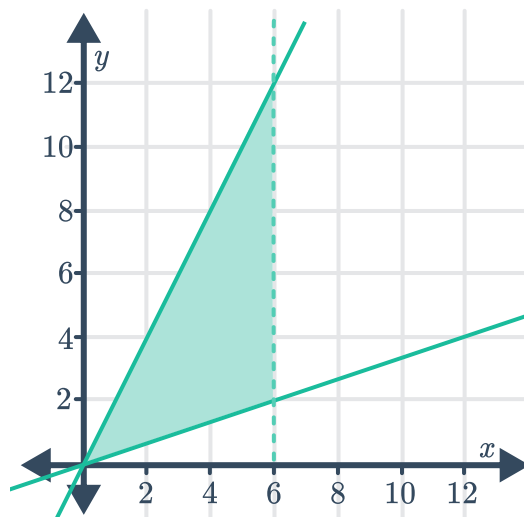
- a Find the area bound by the curve, the x -axis, the y -axis, and the line $x = 3$.
- b Find the equation of the tangent to the curve $y = e^x$ at the point where $x = 3$.
- c Find the exact area enclosed between $y = e^x$, the x -axis, the y -axis, and the tangent at $x = 3$.

Areas between curves

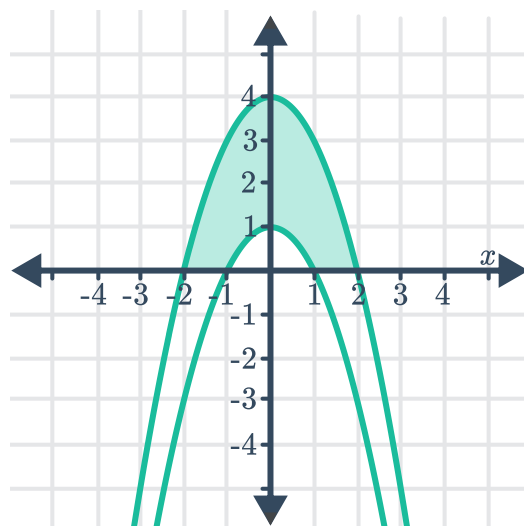


- 14 Find the area enclosed between the lines

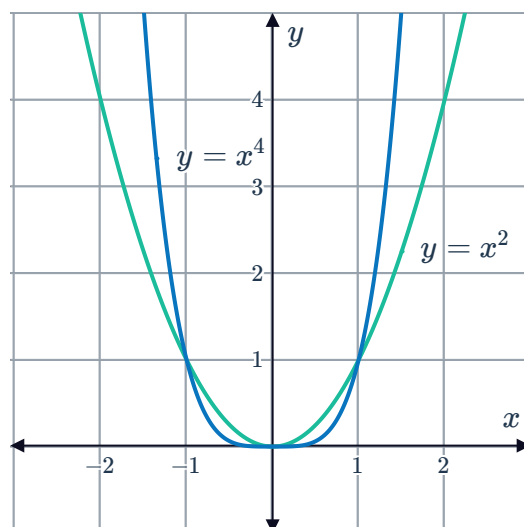
$$y = 2x, y = \frac{1}{3}x \text{ and } x = 6:$$



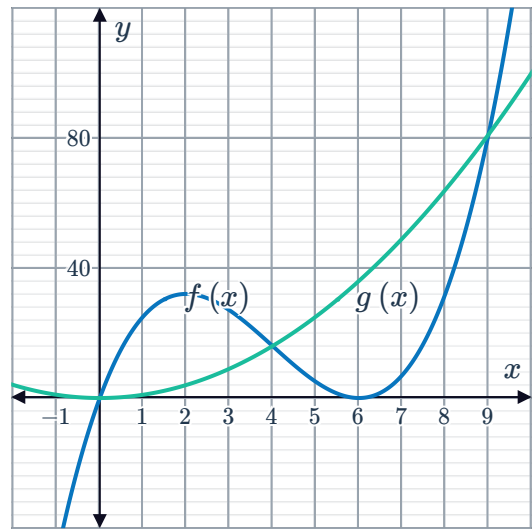
- 15 The diagram shows the shaded region bounded by $y = 4 - x^2$, $y = 1 - x^2$ and the x -axis. Find the area of the shaded region.



- 16 Consider the graph of the functions $y = x^2$ and $y = x^4$. Find the area enclosed between the two curves.



- 17 Consider the graph of the functions $f(x) = x(x-6)^2$ and $g(x) = x^2$:



- State the values of x at which the curves intersect.
- Hence, find the total area bounded between the curves.
- Find the area bounded by the curves and the x -axis.

- 18 For each of the following pairs of functions:

- Graph the functions on the same set of axes.
- State the values of x at which the line and the curve intersect.
- Hence, find the area enclosed between the line and the curve.

a $y = x^2$ and $y = x + 2$

b $y = x(x-4)$ and $y = x$

c $y = -x^2 + 8$ and $y = -x + 2$

d $y = x(x-4)^2$ and $y = x$

e $y = -x(x-3)^2$ and $y = -x$

- 19 For each of the following pairs of functions:

- Find the values of x at which the two curves intersect.
- Find the area enclosed between the two curves.

a $y = x^2 - 48$ and $y = -(x-2)^2 + 4$

b $y = x$ and $y = (x-5)^3 + 5$

c $y = 4x - 12$ and $y = x(x-3)^2$

d $y = 2x$ and $y = x^2 - 15$

- 20 Consider the functions $y = -2x(x-4)$ and $y = -x + 4$.

- Graph the functions on the same set of axes.
- State the values of x at which the curve and the line intersect.
- Hence, find the area enclosed between the curves.
- Find the small area enclosed between the curve, the line and the y -axis.

21 Consider the functions $y = x^2$ and $y = 8 - x^2$ 

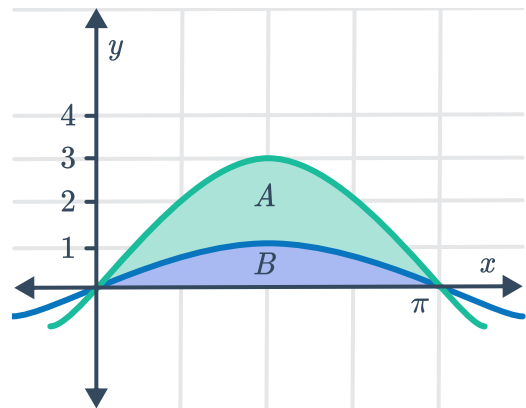
- Graph the functions on the same set of axes.
- State the values of x at which the curves intersect.
- Hence, find the area bounded between the curves.
- Find the small area bounded between the curves and the x -axis, correct to one decimal place.

22 The diagram shows the graphs of $y = \sin x$ and $y = 3 \sin x$:

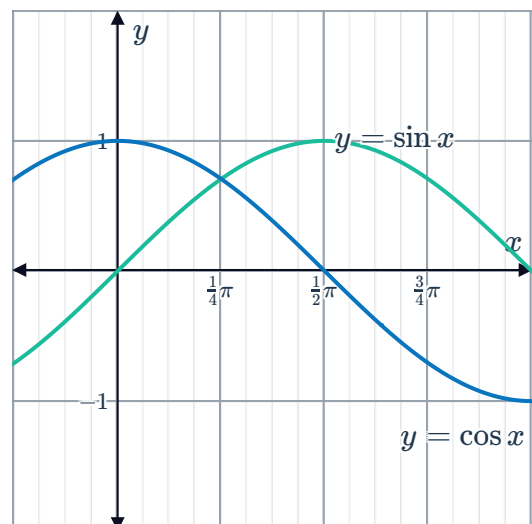
- Determine whether the following integral could be performed to calculate the area of the region bound by the curves, A .

- $\int_0^\pi (3 \sin x - \sin x) dx$
- $\int_0^\pi 3 \sin x dx - \int_0^\pi \sin x dx$
- $\int_0^\pi 3 \sin x dx + \int_0^\pi \sin x dx$
- $2 \int_0^\pi \sin x dx$
- $\int_0^\pi (3 \sin x + \sin x) dx$
- $4 \int_0^\pi \sin x dx$

- Find the area of the shaded region A .
- What is the ratio of the areas $A : B$?



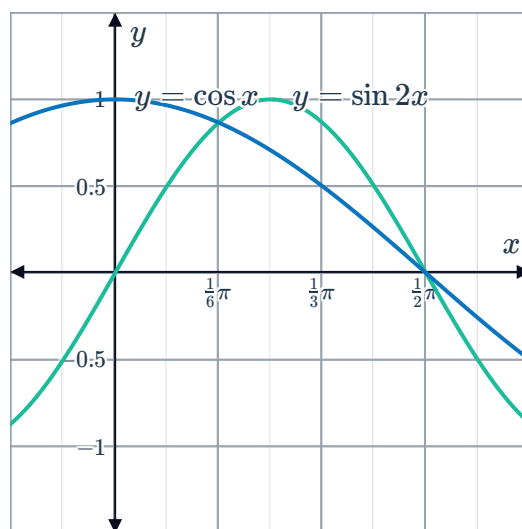
23 Find the area of the region bounded by the curves $y = \sin x$ and $y = \cos x$, and the lines $x = 0$ and $x = \frac{\pi}{4}$:



24 Find the area enclosed by the two curves

$y = \sin 2x$ and $y = \cos x$ on the interval

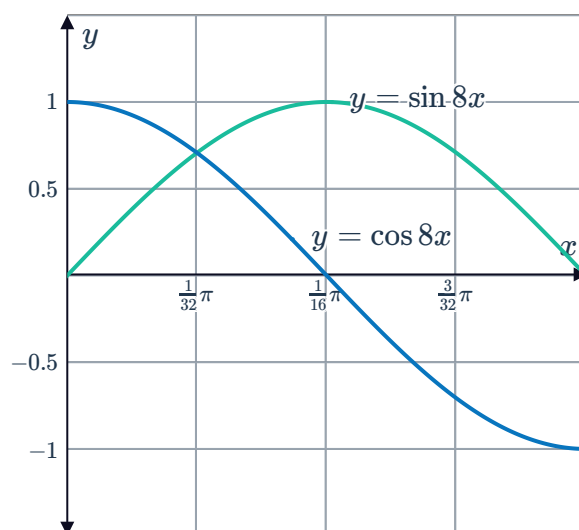
$$\frac{\pi}{6} \leq x \leq \frac{\pi}{2}.$$



25 Find the area between the two curves of

$y = \sin 8x$ and $y = \cos 8x$ on the interval

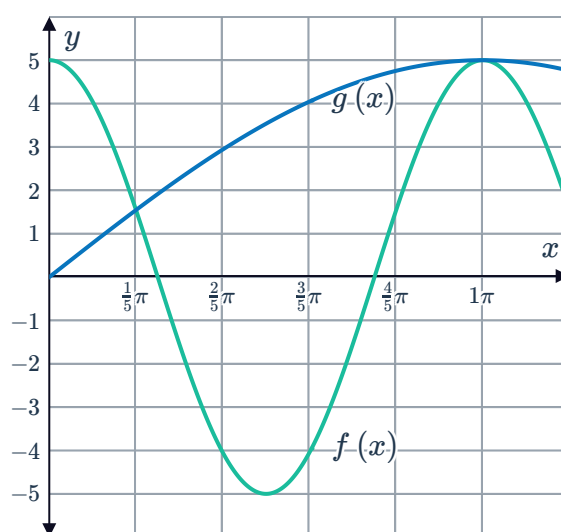
$$\left[0, \frac{\pi}{16}\right].$$



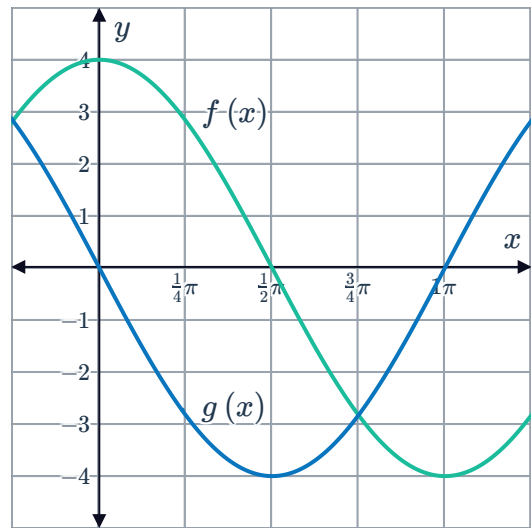
26 Find the area enclosed by the two curves of

$$f(x) = 5 \cos 2x \text{ and } g(x) = 5 \sin \left(\frac{x}{2}\right)$$

between $x = \frac{\pi}{5}$ and $x = \pi$. Round your answer to two decimal places.



- 27** Find the area of the region bounded by $f(x) = 4 \cos x$, $g(x) = -4 \cos\left(x - \frac{\pi}{2}\right)$, the y -axis and $x = \frac{3\pi}{4}$.



- 28** Find the area of the region bounded by the line $y = 2\pi - x$ and the curve $y = -2 \sin x$ and the x -axis between $x = 0$ and $x = \frac{2\pi}{3}$.

- 29** Consider the functions $y = \sin x$ and $y = \cos x$.

- Graph the functions on the same set of axes in the interval $[0, \pi]$.
- Hence, find the area between the two curves in this interval.

- 30** Consider the functions $y = \sin 4x$ and $y = \cos 4x$ for $0 \leq x \leq \frac{\pi}{2}$.

- Find the x -coordinates of the points of intersection of the two curves in the given interval.
- Graph the two functions on the same set of axes.
- Hence, find the area bounded by the two curves between the points of intersection found in part (a).

- 31** Consider the functions $f(x) = \sin 4x$ and $g(x) = 1 - \cos 4x$.

- Graph the functions from 0 to $\frac{\pi}{2}$ on the same set of axes.
- For what values of x is $f(x) = g(x)$?
- Evaluate $\int_0^{\frac{\pi}{2}} (1 - \cos 4x - \sin 4x) dx$.
- Find the area between the two curves on the interval $\left[0, \frac{\pi}{2}\right]$.

- 32** Find the exact area between the following graphs and lines:

- | | |
|--|---|
| a $y = x^3 - 8$, $x = 2$, and $y = -8$ | b $y = e^x$, $y = e^{-x}$, and $x = 3$ |
| c $y = e^{5x}$, $y = e^{-5x}$, $x = -2$, and $x = 2$ | d $y = x$, $y = e^{1-x}$, and $x = 3$ |

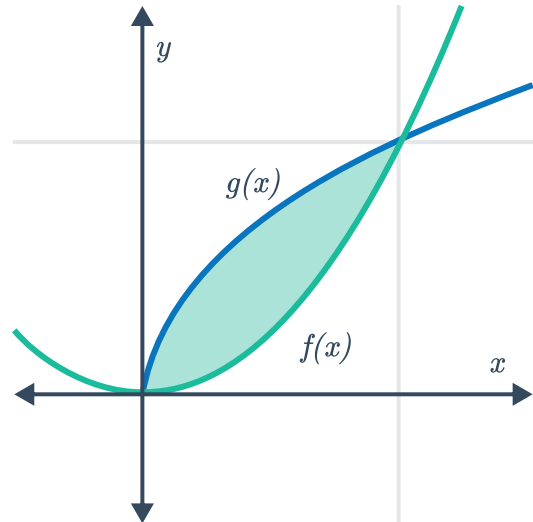
33 Consider the functions $f(x) = e^x + 12$ and $g(x) = 4x + e^3$.

- Find the x -coordinate of the point of intersection that is to the right of origin.
- Find the area in the first quadrant bounded by the two curves and the y -axis.

34 Consider the graphs of $f(x) = x^2$ and $g(x) = \sqrt{x}$:

- Find the x -coordinates of the points of intersection.
- Find the shaded area bounded by the two curves.
- Complete the following table for the area bounded by $f(x) = x^n$ and $g(x) = \sqrt[n]{x}$.

n	2	3	5	8
Area				

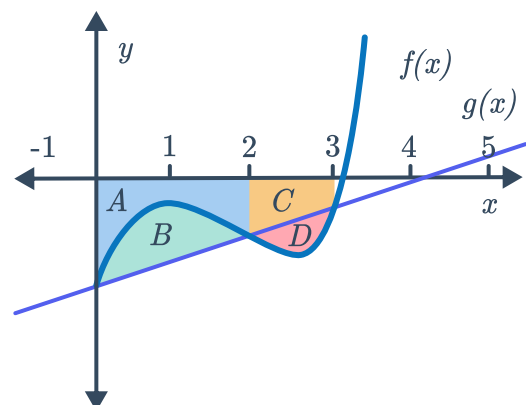


- Make a conjecture for the value of the area between the curves $f(x) = x^n$ and $g(x) = \sqrt[n]{x}$ for positive integer values of n .
- Prove your conjecture.

35 Consider the graph below which has the following properties:

- Functions $f(x)$ and $g(x)$ intersect at $x = 0$, $x = 2$ and $x = 3$
- Region A has an area of 22 square units
- Regions A and B have a combined area of 56 square units
- $\int_2^3 f(x) dx = -18$ and $\int_2^3 g(x) dx = -13$

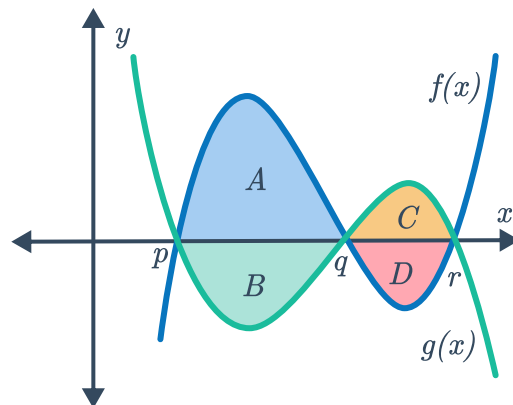
- Find $\int_0^2 f(x) dx$.
- Find $\int_0^2 (f(x) - g(x)) dx$.
- Find the area of region C .
- Find the area of region D .
- Find $\int_0^3 (f(x) - g(x)) dx$.



36 Consider the graph below which has the following properties:

- Functions $f(x)$ and $g(x)$ intersect on the x -axis at points p, q , and r
- Region A has an area of 16 units^2
- Regions A and B have a combined area of 26 units^2
- $\int_p^r f(x) dx = 11$ and $\int_p^r g(x) dx = -9$

- Find the value of $\int_p^q f(x) dx$.
- Find the value of $\int_p^q g(x) dx$.
- Find the area of region C .
- Find the value of $\int_p^r (f(x) - g(x)) dx$.



37 Consider the graph of the functions $f(x) = 3x$ and $g(x) = \frac{x^2}{2}$ together with the line $x = 8$:

Region P is the area enclosed by f and g .
Region Q is the area enclosed by f and g and $x = 8$.

- Find the area of region P .
- Find the area of region Q .
- $f(x)$ is redefined such that $f(x) = ax$ and the area of region P is half the area of region Q . Calculate the value of a that makes the statement true.

